

Python 3.15 Data Engineering Readiness Assessment

Overall Readiness Score: 87% # Python 3.15 Data Engineering Readiness Matrix

Project: Python 3.15 Data Engineering Validation Suite **Author:** Dr. Ceasar Jackson Jr. **Python Version:** 3.15.0b2 **Assessment Date:** 2026-06-07

Overall Readiness

Area	Status
Runtime	PASS
Analytics	PASS
Databases	PASS
Visualization	PASS
Orchestration	PARTIAL
Distributed Processing	PARTIAL
Arrow Ecosystem	BLOCKED

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Core Stack

Package	Version	Status
NumPy	2.4.6	PASS
Pandas	3.0.3	PASS
Polars	1.41.2	PASS
DuckDB	1.5.3	PASS
SQLAlchemy	2.0.50	PASS
Pydantic	2.13.4	PASS
Matplotlib	3.10.9	PASS
Plotly	6.7.0	PASS

Extended Stack

Package	Status	Notes
MLflow	PASS	Experiment tracking validated
Apache Airflow	PASS	DAG validation successful
Dask	INCOMPAT	PyArrow dependency blocker
Prefect	INCOMPAT	Python 3.15 typing API change
PySpark	SKIP	Docker image not available
Ray	SKIP	Not installed
Delta Lake	SKIP	Not installed

Known Blockers

PyArrow

- No CPython 3.15 wheels available.
- Prevents Dask DataFrame support.
- Blocks portions of the Arrow ecosystem.

Prefect

- References removed typing APIs.
- Requires upstream Python 3.15 compatibility update.

Recommendation

Python 3.15 is suitable for advanced testing and validation environments.

Current Status: READY FOR ADVANCED TESTING

Production Target: Python 3.15 Release Candidate